

Accounting for Love of Variety in Consumer Demand Estimation

Tate Mason

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1 Introduction

In the baseline logit model, a consumer is perceived as having a fixed preference for each product, and the utility from consuming a product is independent of past consumption. Further, it is assumed that a new good is always a utility boosting event, by the error term's property of being identically and independently distributed. However, in many markets, consumers may have a love of variety, or a desire to consume different products over time. This can be due to factors such as boredom, satiation, or a preference for novelty. In such cases, the utility from consuming a product may depend on past consumption, and the introduction of a new good may not always lead to an increase in utility.

I find that...

The rest of the paper is organized as follows: section two discusses the related literature and section three describes the data used and some descriptive statistics. Section four showcases the model and environment and section five presents results. Section six goes into counterfactual analysis, while section 7 concludes.

2 Related Literature

The variety-seeking and state dependence literature has been primarily focused on understanding the dynamics of consumer preferences and demand over time. From ?, who developed an empirical structural model of state dependence in consumer choice, to ?, who iterated the model to allow for consumer learning. ? is the most closely related paper, as he develops a model of variety-seeking behavior which is both tractible and flexible, and uses Nielsen data to estimate the model. However, this paper only looks at demand, leaving me a niche to explore the interplay between variety-seeking and strategic advertising.

3 Data

3.1 Data

My data comes from Nielsen’s HomeScan and MarketScan data. These two main datasets give me access to household purchase decisions as well as product characteristics. I am looking at the years between 2014-2016 in markets Atlanta, Chicago, San Diego, and Denver. Households are comprised of a single-person to avoid any contamination of shopping for multiple people like children or spouses, allowing me a cleaner sample of shopping trips. These trips are encoded with unique identifiers, thus allowing me to look at per-trip, and within-trip, decisions. Stores have to be valid Nielsen stores, meaning they appear in the reatil code data. This is necessary to observe prices the products were bought at. The product space I am looking at is yogurt. Yogurt is good for this sort of analysis due to its relative lack of storability as well as its variation in attributes like flavor, nutrient contents, and types like Greek or Skyr. I have limited yogurt sizes to cups. Cups of yogurt are between five and eight ounces and are single serving products which come in more varied flavors.

I classify flavors into three categories: berry, plain, and other. While these are not taking full advantage of the granularity in flavors available, the classifications capture the two largest categories, berry and plain. I define the outside option as a trip in which yogurt was not ever purchased. 1 shows that the outside option is taken 80% of the time each trip.

4 Model

4.1 Environment

There are a continuum of $i \in \mathcal{I}$ consumers who exist for $\mathcal{T}$ periods. Within a given period, a household can go for a shopping trip $t \in \mathcal{T}$ which is comprised of choice occasions $o \in \mathcal{O}_t$. Each choice occasion allows for the purchase of a singular good.

4.2 Consumer’s Problem

The consumer’s problem is the following. In each period t , the consumer i chooses a product j from the menu of available products J_t . Each product $j \in J_t$ is a bundle of $k \in K$ characteristics such that $X_{jt}^k \in \mathcal{X}_{jt}$. The consumer’s utility from choosing product j at time t is given by:

β_i^k is a consumer’s utility from characteristic X_{jt}^k , γ_i is their preference for variety, Ξ_{ijt} is the difference between the current characteristic and the previous choices, θ_{it}^k . θ_{it}^k is a function of the consumers’ past purchases, where λ weights chosen products in the past, X_{js}^{*k} and the most recently

chosen product, X_{jt-1}^{*k} , allowing for more recent purchases to play more of a role in the current period's decision. This could take many forms, but in this paper will be the mean characteristic of past purchases. α_i is a heterogeneous price sensitivity to p_{jt} . ξ_j is unobserved quality in product j , and ε_{ijt} is the Gumbel shock which allows for a closed form logit.

The choice probability of consumer i choosing product j in time t is as follows:

$$P_{ijt} = \frac{\exp U_{ijt}}{\sum_k \exp U_{ikt}}$$

5 Results

6 Counterfactuals

7 Conclusion

In closing, this paper develops a dynamic discrete choice model of consumer demand that incorporates a love of variety and strategic advertising by firms. The model is estimated using scanner data, and counterfactual analyses are conducted to evaluate the effects of different market structures on consumer welfare and market outcomes. The paper contributes to the literature on advertising and consumer behavior, by providing empirical evidence on the effects of strategic advertising in a dynamic demand setting. The findings of this paper have important implications for policymakers and firms, as they highlight the potential benefits and drawbacks of targeted advertising in markets when accounting for consumers' preference for variety.

8 Tables and Figures

Table 1: Summary Statistics for Household Panel, Switching Dynamics, and Pricing

Statistic / Variable	Value	Unit / Description
Panel & Household Demographics		
Total Households in Sample	1,269	Households
Yogurt Purchasing Households	715	Households (56%)
Mean Trips per Household	17.65	Trips / Household
Mean Purchases per Shopping Trip	5.91	Units / Trip
Mean Household Income	17.62	Income Bracket
Median Household Income	18.00	Income Bracket
Outside Option Rate	80%	Non-purchase trip share
Racial Composition of Households		
White (Code 1)	955	Households (75%)
Black (Code 2)	254	Households (20%)
Asian (Code 3)	24	Households (2%)
Other / Multiracial (Code 4)	36	Households (3%)
Switching & Purchasing Dynamics		
Ever-Switcher Households	35%	Conditional on yogurt purchase
Coupon Purchase Rate (All Purchases)	48%	Share of yogurt trips
Switching Attributable to Coupon	17%	Share of switch events
Mean Times Switching per Household	8.60	Switches / Household
Mean Consecutive Buys per Flavor	10.16	Continuous purchase trips
Mean Spell Duration per Flavor	10.16	Trips per flavor spell
Yogurt Pricing		
Overall Mean Price	\$0.84	Per unit price
Other Flavors Price	\$0.76	Per unit price
Berry Price	\$0.92	Per unit price
Plain Price	\$1.16	Per unit price

